

$$\lim_{x \rightarrow c} f(x) =$$



$$\sqrt{\sin^2 x}$$
$$\int_0^f$$

# CALCULUS

## 1/2 REVIEW

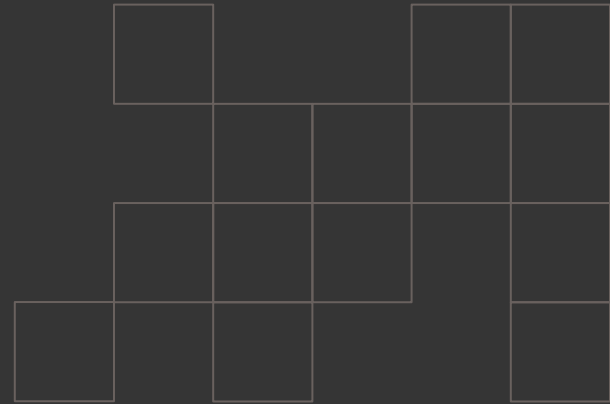
$$\frac{d}{dx} \sin(x) = \cos(x)$$
$$\sin^2 x + \cos^2 x = 1$$

$$\frac{d}{dx} \sin(x)$$



# Calculus 1/2 Final Exam Review

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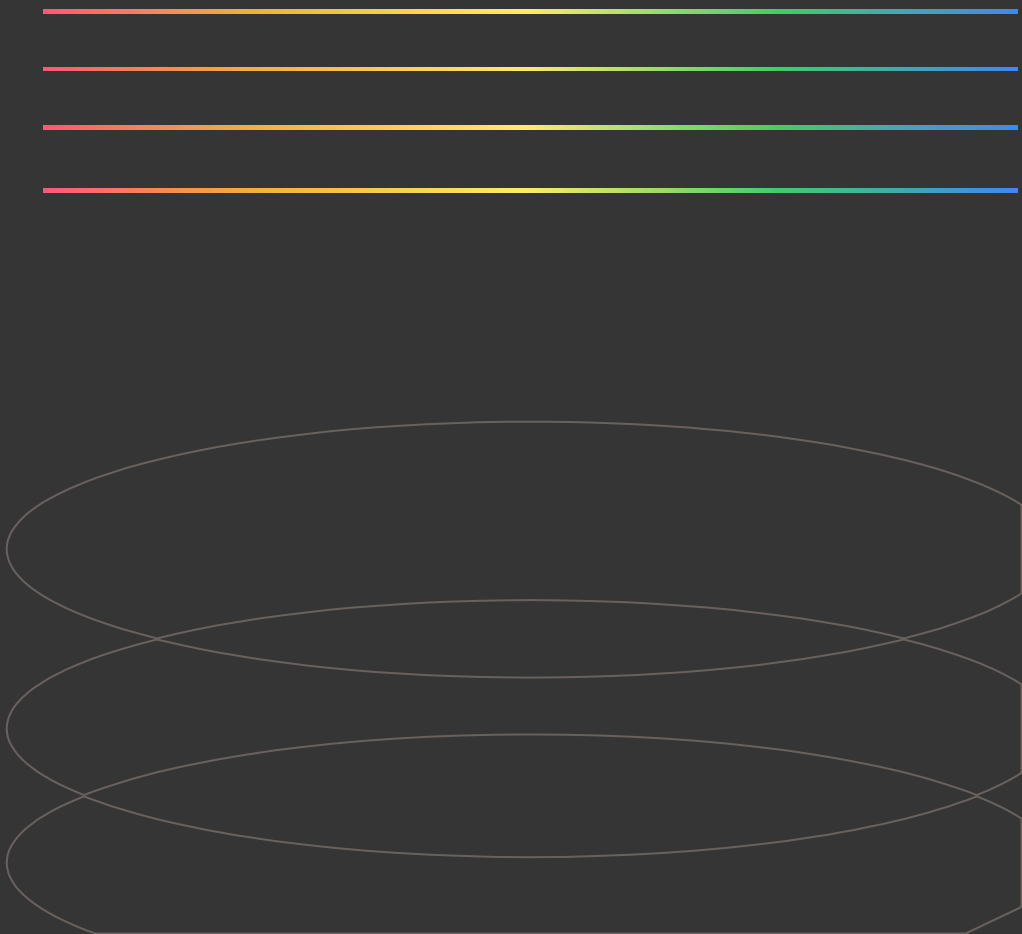
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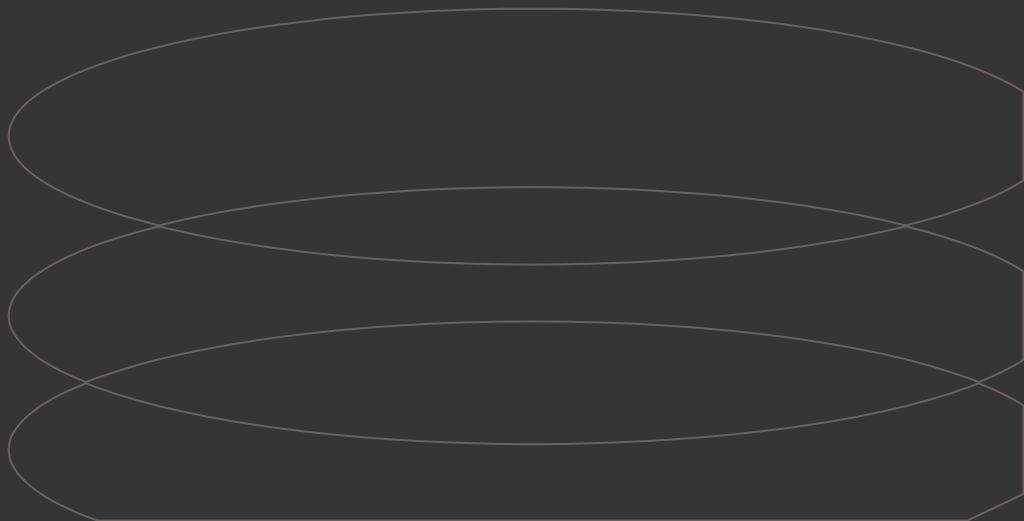
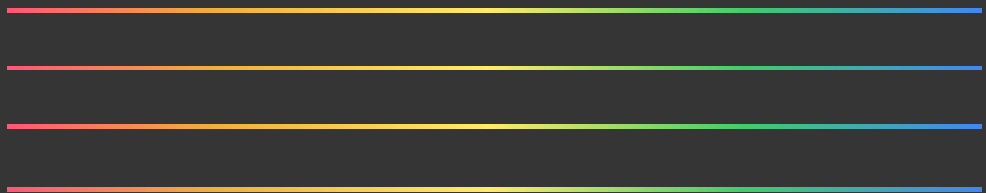
# Limits

- REMEMBER the basic rules
- UNDERSTAND epsilon-delta
- REVIEW old examples
- SOLVE hard questions



# Hard Limits

- Here follow some harder questions, mainly L'Hopital.
- Try these out, work with a partner.



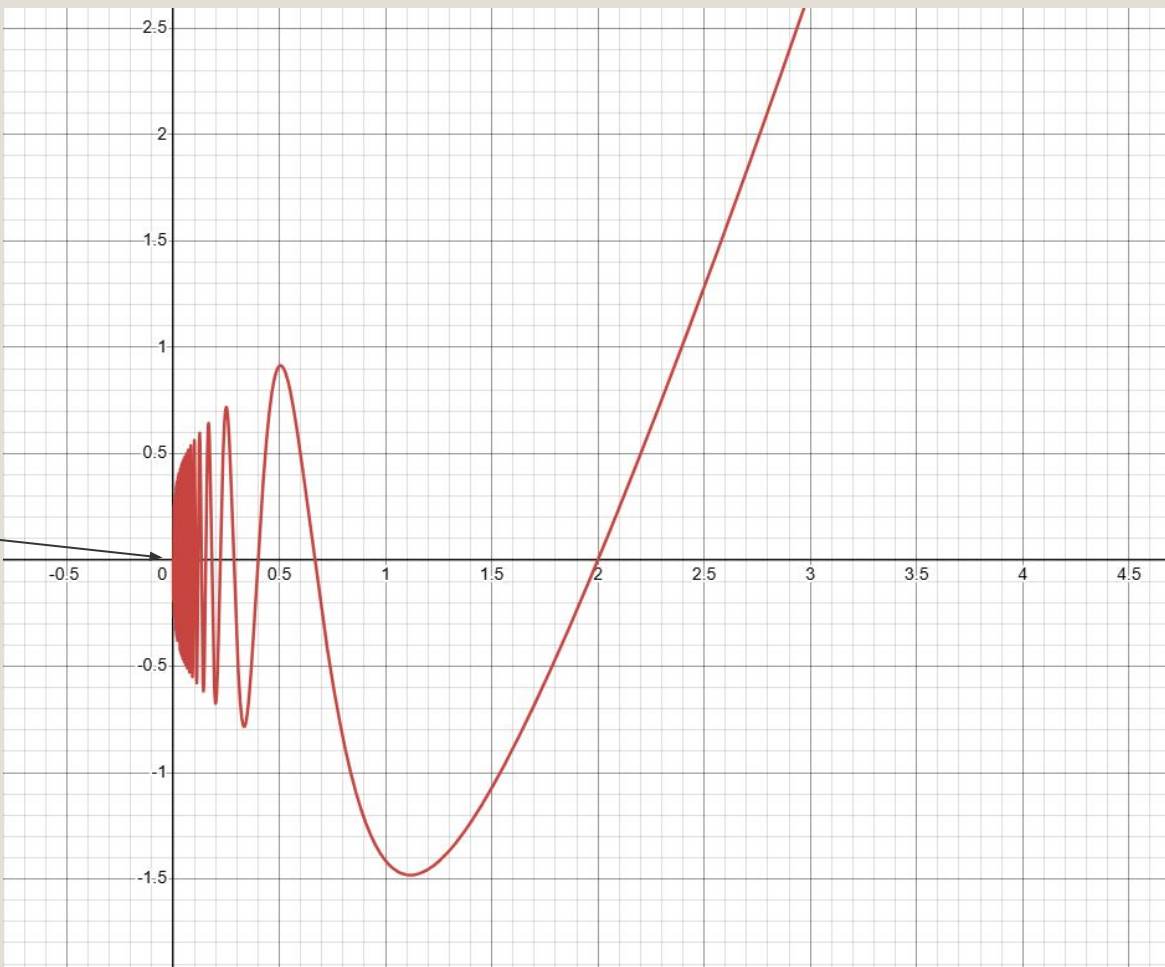
2. Evaluate the following limit. (For those that have had calculus before, feel free to confirm your answer using L'Hôpital's rule, but solve it another way. This is an algebra challenge!)

$$\lim_{x \rightarrow 64} \frac{\sqrt[3]{x} - 4}{\sqrt{x} - 8}$$

$$\lim_{x \rightarrow 0} \sqrt{\sqrt{x} + x^3} \cos \frac{\pi}{x}$$

$$\lim_{x \rightarrow 0} \sqrt{\sqrt{x} + x^3} \cos \frac{\pi}{x}$$

Not so  
nefarious!



$$\lim_{x \rightarrow \infty} x^{1/(\ln x)}$$

$$\lim_{x \rightarrow 0} (\cos x)^{\csc x}$$

$$\lim_{x \rightarrow \pi/2^-} \cos x \ln \tan x$$

$$\lim_{x \rightarrow \infty} x^{\sin(1/x)}$$

$$\lim_{x \rightarrow \infty} x^{1/(\ln x)}$$

*Ans.*  $e$ .

$$\lim_{x \rightarrow 0} (\cos x)^{\csc x}$$

*Ans.*  $1$ .

$$\lim_{x \rightarrow \pi/2^-} \cos x \ln \tan x$$

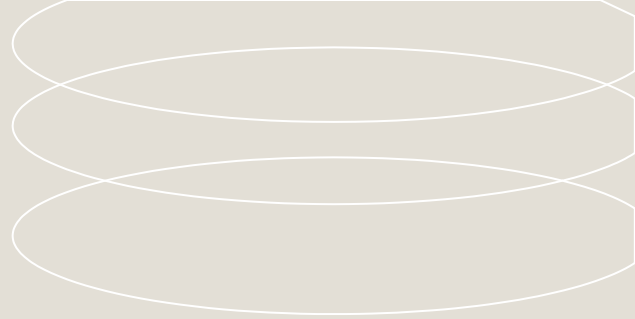
*Ans.*  $0$ .  
[SSJ 124.17]

$$\lim_{x \rightarrow \infty} x^{\sin(1/x)}$$

*Ans.*  $1$ .

$$\lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x \cos^{-1} x}$$

Just try what you know!



$$\star 57. \lim_{x \rightarrow 0^+} \left( \frac{\sin x}{x} \right)^{1/x^2}$$

$$\star 58. \lim_{x \rightarrow 0} \left( \frac{1}{\sin^2 x} - \frac{1}{x^2} \right)$$

$$\star\star 59. \lim_{x \rightarrow +\infty} \frac{x^{\sqrt{\ln x}} (\sqrt{\ln x})^x}{(\sqrt{x})^{\ln x} (\ln x)^{\sqrt{x}}}$$

$$\star 57. \quad \lim_{x \rightarrow 0^+} \left( \frac{\sin x}{x} \right)^{1/x^2}$$

*Ans.*  $e^{-1/6}$ .  
[R 4.7.75(a)]

$$\star 58. \quad \lim_{x \rightarrow 0} \left( \frac{1}{\sin^2 x} - \frac{1}{x^2} \right)$$

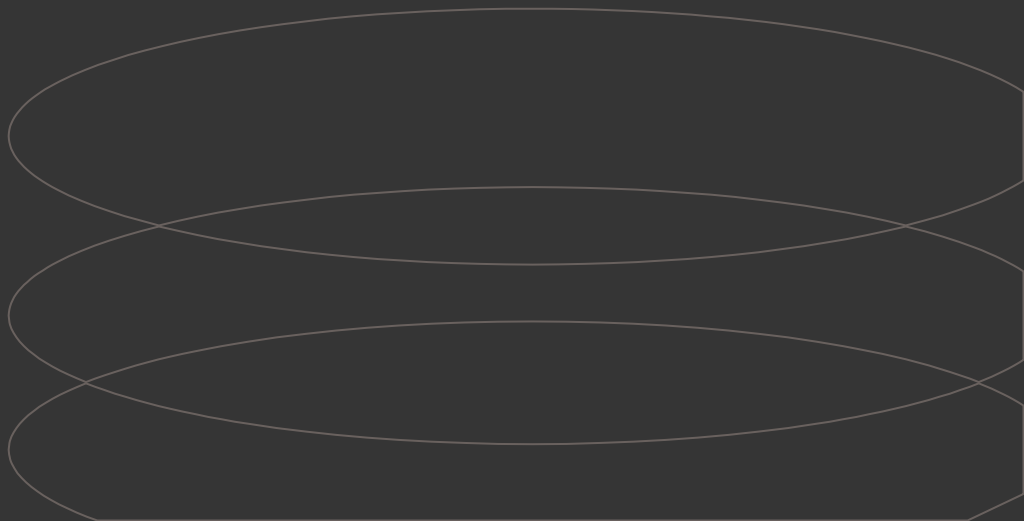
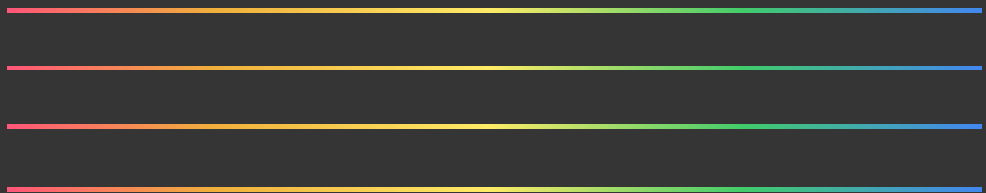
*Ans.*  $\frac{1}{3}$ .  
[R 4.7.75(b)]

$$\star\star 59. \quad \lim_{x \rightarrow +\infty} \frac{x^{\sqrt{\ln x}} (\sqrt{\ln x})^x}{(\sqrt{x})^{\ln x} (\ln x)^{\sqrt{x}}}$$

*Ans.*  $\infty$ .  
[W VIII.4.11]

# Derivatives

- Remember the key rules; product, quotient, chain, etc.
- Understand implicit differentiation, etc. to apply derivatives



You should be familiar with implicit/logarithmic differentiation; use this to figure out the derivatives of:

- $x^x$  (should be review)
- $x^{\sin(x)}$
- $x^{(x^x)}$
- Use the first derivative you figured out, it's ok

**Example 1** Differentiate the function.

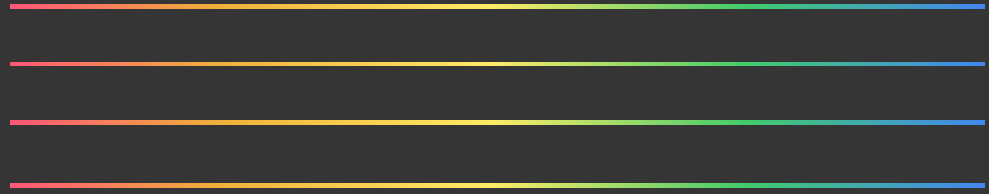
$$y = \frac{x^5}{(1 - 10x) \sqrt{x^2 + 2}}$$

$$\frac{\sqrt{5t+8} \sqrt[3]{1-9\cos(4t)}}{\sqrt[4]{t^2+10t}} \left[ \frac{\frac{5}{2}}{5t+8} + \frac{12\sin(4t)}{1-9\cos(4t)} - \frac{\frac{1}{2}t + \frac{5}{2}}{t^2+10t} \right]$$

Find an expression for the  
following:

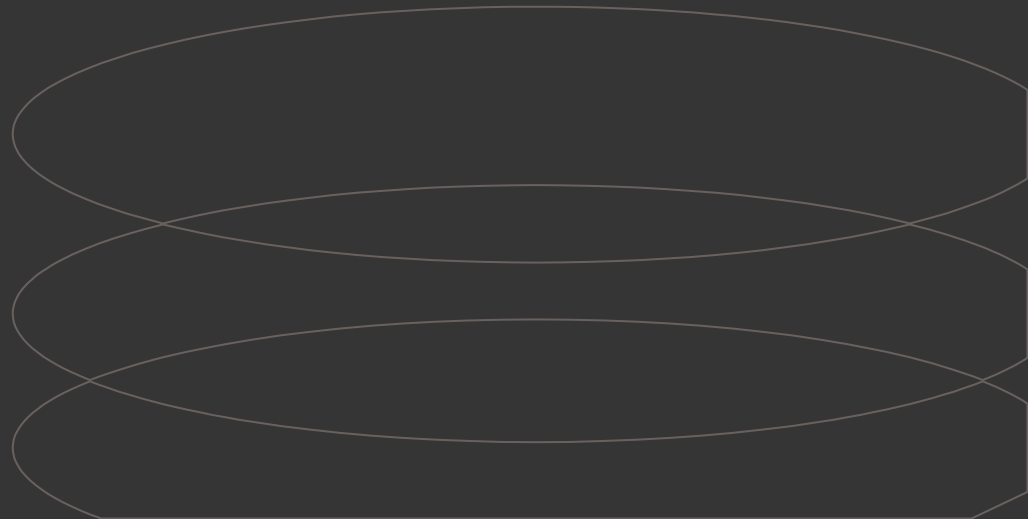
$$f^{(n)}(x), f(x) = xe^x$$

# Applications of Derivatives



Most important thing is to understand the context of a derivative and what it really means.

As Tr. Bellotti says, “Context is king!”



. A spherical ball is being inflated at a constant rate of  $10 \text{ in}^3/\text{s}$ .  
When the surface area is  $100\pi \text{ in}^2$ , how fast is the radius of the ball increasing?

A 25-foot ladder leans against a vertical wall. Both its base and top are a whole number of feet away from the bottom of the wall. If the ladder's angle with the ground is greater than  $45^\circ$  and its base is sliding away from the wall at 2 feet per second, find the sum of the possible speeds at which the top of the ladder can slide down.

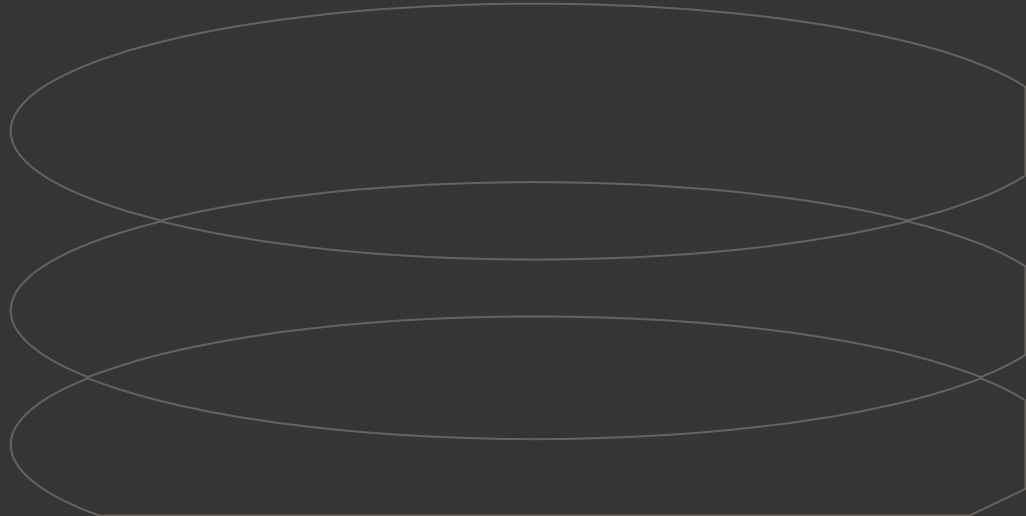
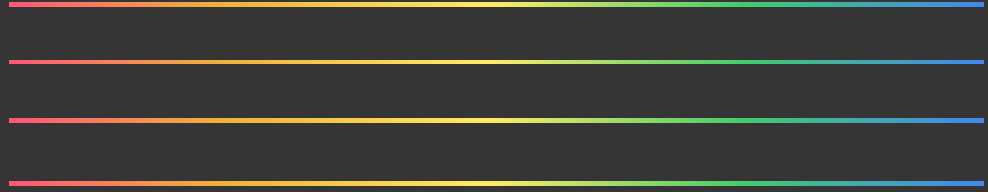
A  $a$ -foot giant lumbers away from a lamp post  $b$  feet tall at the speed of  $c$  feet per second. How fast is his shadow moving?

For  $x$ -values near  $60^\circ$ ,  $\sin(x) \sim \frac{180\sqrt{3} + \pi(x-60)}{360}$ . Explain why this is so and find a similar estimation for  $x$ -values near  $45^\circ$ .

# Optimization

Move everything into one variable and then try to optimize it.

Make sure to check the context of the problem to find what you're being asked, check for sensibility, etc.



. What are the relative maximum and minimum values of the cubic  $y = x^3 + 10.5x^2 + 36x - 3$ ?

What is the shortest distance between two relative extrema on the graph of  $y = \frac{4x^2}{e^x}$ ?

Which cylinder of a given volume has the least overall surface?

It is required to make an open rectangular box of greatest capacity out of a square sheet of cardboard with side  $a$  by cutting squares at each of the angles and bending up the ends of the resulting cross-like figure.

Inscribe in a given ellipse a rectangle of largest area with sides parallel to the axes of the ellipse.

$$\int \frac{1}{1 + \sin x} dx$$

$$\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$$

$$\int \frac{\sin(4x) \cos(x)}{\cos(2x) \sin(x)} dx$$

$$\int_0^1 x^{2024} (1 - x^{2025})^{2025} dx$$

$$\int_1^{e^e} \frac{\log \left( x^{\log(x^x)} \right)}{x^2} dx$$

$$\int_0^{2\pi} \sin(x) \cos(x) \tan(x) \cot(x) \sec(x) \csc(x) \, dx$$

$$\int \frac{dx}{x \log x + 2x}$$

$$\int \frac{x^{2024} - 1}{x^{506} - 1} dx$$